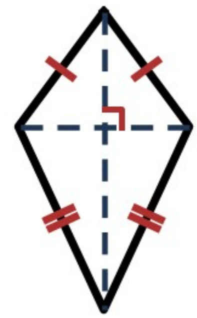
Speak **L**ike **A** **M**athematician:

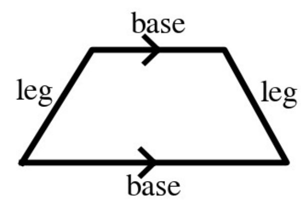
kite:

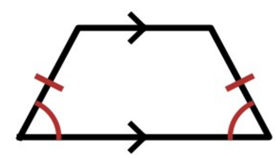
trapezoid · base · leg:

base angles · isosceles trapezoid:

midsegment of a trapezoid:







Kite and Trapezoid Theorems

Kite: the diagonals are ____________________, and exactly ONE pair of opposite angles is congruent — the angles between the NON-congruent sides.

Isosceles trapezoid: each pair of base angles is ____________________, and the diagonals are ____________________ (both are if-and-only-if).

Trapezoid Midsegment Theorem: the midsegment is ____________________ to both bases, and its length is ____________________ the SUM of the bases.

Example 1 – Kite Angles

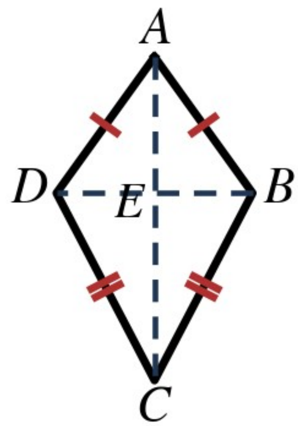
In kite ABCD at the right, $\overline{AB} \cong \overline{AD}$, $\overline{CB} \cong \overline{CD}$, m∠BAD = 48°, and m∠ABC = 121°.

a) Which angles must be congruent? Why?

b) Find m∠ADC.

c) Find m∠BCD.

d) What is m∠AEB, where the diagonals meet at E?

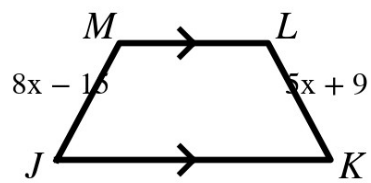


Example 2 – Make It Isosceles

In trapezoid JKLM at the right, $\overline{JM}$ and $\overline{KL}$ are the legs, JM = 8x − 15, and KL = 5x + 9.

a) Find x so that JKLM is isosceles.

b) Find the length of each leg.

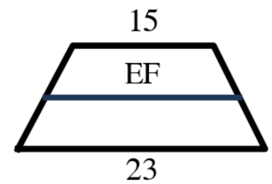


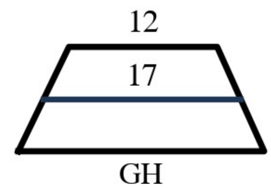
Example 3 – Use the Midsegment Theorem

Each trapezoid at the right shows its midsegment.

a) Find EF in the first figure (bases 23 and 15).

b) Find GH in the second figure (midsegment 17, one base 12).



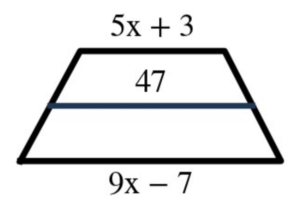


Example 4 – Midsegment Algebra

In the trapezoid at the right, the bases measure 9x − 7 and 5x + 3, and the midsegment measures 47.

a) Write the equation and solve for x.

b) Find both bases and check.



Example 5 – Trapezoid Angles

In isosceles trapezoid PQRS at the right, $\overline{PQ} \parallel \overline{SR}$ and m∠P = 68°.

a) Find m∠S.

b) Find m∠Q and m∠R.